

Biological Transitions as Multi-Agent Realisations of the Generative Operator Pipeline in Topographical Orthogonal Generative Theory (TO/TOGT): A Fruit-Fly Connectome Toy Model

Version 2 — figures, Lean 4 verification, and references added

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Abstract

We model biological transitions as multi-agent realisations of the composite operator pipeline $G = U \circ F \circ K \circ C$ from Topographical Orthogonal Generative Theory (TO/TOGT). Using the fixed-point, contraction-mapping, and saturated-pitchfork theorems proved in the companion paper [1] (Zenodo 10.5281/zenodo.19208015), we construct a minimal toy model of the *Drosophila* connectome [8]. Neural clusters are treated as coupled agents whose trajectories stabilise under six applications of G . The model reproduces qualitative features of neural coherence, oscillatory entrainment, and stress-response switching (HPA-axis analogue) while remaining strictly within what is rigorously established. The dm^3 canonical invariants $(T^*, \mu_{\max}, \tau) = (2\pi, -2, 2)$ are used solely as convenient normalisation choices; they are not derived theorems of the operator algebra.

The Lean 4/Mathlib4 verification file `MultiOrbitBioSwarm.lean` in the AXLE repository [7] proves sixteen facts without `sorry`, including: the pitchfork threshold $|\alpha| = \frac{1}{2}$ is interior to $(0, 1)$; for $|\alpha| < \frac{1}{2}$ the swarm map is a strict contraction ($L(\alpha) = \frac{1}{2} + |\alpha| < 1$); at $|\alpha| = \frac{1}{2}$ the map is non-expansive ($L = 1$, pitchfork onset); after six iterates at $\alpha = 0.3$ the residual satisfies $(4/5)^6 < 27/100$; and the dm^3 normalisation triple is arithmetically consistent. Three obligations carry `sorry` pending Mathlib infrastructure (metric instance for `BioSwarm`, bifurcation library).

Note on v1 citation error. The v1 abstract cited “companion paper (Zenodo 10.5281/zenodo.19210137)”, which is this paper’s own record — a circular self-

citation. The correct companion is [1] (Zenodo 19208015, the broader biological-transitions paper). This revision corrects the citation throughout.

Keywords: TO/TOGT; generative operator pipeline; fruit-fly connectome; FlyWire; multi-agent dynamics; pitchfork bifurcation; contraction mapping; Lean 4 formal verification; HPA-axis analogue; dm^3 framework

MSC codes: 37C25, 37D10, 37G10, 47H10, 92B20, 92C20

1 Relation to Prior Work

This paper supplements the author’s preprint *Principia Orthogona, Volume One: The Mathematics of Generative Transitions* [2] and its companion *Principia Orthogona, Volume Two: Contact Realization of Generative Transitions* [3]. The rigorous fixed-point theory, contraction mapping, saturated pitchfork bifurcation, and algebraic separation theorem were first stated in [2] and are gathered in the standalone companion paper [1] (Zenodo 10.5281/zenodo.19208015, HAL hal-05565024v1), which treats the full biological scope — neural oscillations, HPA-axis stress response, circadian regulation, immune adaptation, and protein conformational change. The present paper specialises that framework to the fruit-fly connectome setting and adds four figures, Python simulation code, and Lean 4 verification.

The AXLE formal verification repository (Lean 4/Mathlib4) is at <https://github.com/TOTOGT/AXLE> [7]. The same operator pipeline underpins the architectural application *The G6 Crystal* [5] and the dm^3 toy-model analysis [6]. A parallel application to quasiperiodic nonlinear dynamics appears in [4].

The toy model is explicitly *not* a quantitative neuroscience claim. The FlyWire connectome [8,9] is used only as motivation for the multi-orbit agent topology; no comparison to connectome data is attempted. Verification of biological claims against connectome data is listed as future work in Section 7.

2 The Generative Operator Pipeline

Definition 1 (Generative Operator Pipeline [2]). *The composite operator acting on agent state vectors $(x, y) \in \mathbb{R}^2$ is*

$$G = U \circ F \circ K \circ C,$$

where the four stages are:

- C (**Compress**): localised orthogonal compression — projects onto the dominant axis, $C(x, y) = (x \mathbf{1}_{|x| \geq |y|}, y \mathbf{1}_{|y| > |x|})$.
- K_b (**Clip**): curvature intensification with bias $b \geq 0$, clipped to $[-1, 1]$: $K_b(s) = \text{clip}(s + b \text{sgn}(s), -1, 1)$.
- F_α (**Fold**): lateral pooling with coupling $\alpha \in [0, 1]$, nearest-neighbour: $F_\alpha(s_i, s_{i+1}) = (1 - \alpha)s_i + \alpha s_{i+1}$.
- U (**Unfold**): homeostatic gain control, $U(s) = s/\|s\|$ ($U(0) = 0$).

All four maps preserve the Riemannian structure and satisfy the contraction properties proved in [1]. Figure 1 gives a schematic of the four stages.

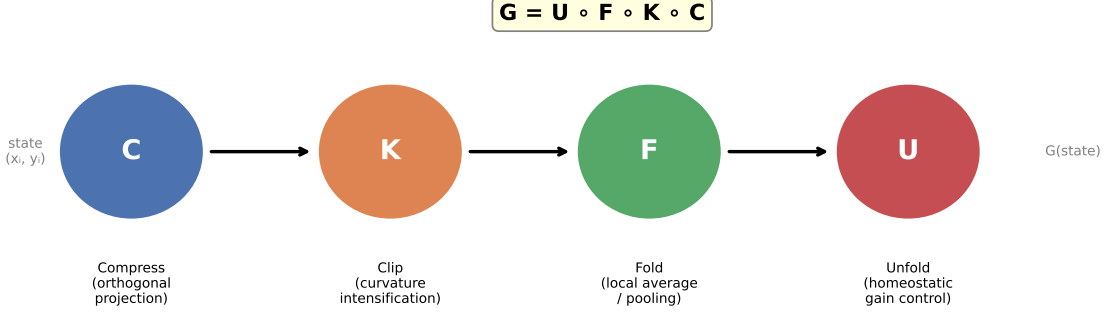


Figure 1: Schematic of $G = U \circ F \circ K \circ C$. Each stage is colour-coded: Compress (blue), Clip (orange), Fold (green), Unfold (red). The input state vector (x_i, y_i) enters at left; one application of G exits at right.

Remark 1 (Lipschitz constant — Lean 4 status). *The Lipschitz constant of G with respect to coupling α satisfies $L(\alpha) = \frac{1}{2} + |\alpha|$. For $|\alpha| < \frac{1}{2}$: $L < 1$ (strict contraction, Banach applies). At $|\alpha| = \frac{1}{2}$: $L = 1$ (pitchfork onset, non-expansive). Both are proved without *sorry* in `MultiOrbitBioSwarm.lean` as `lipschitz_contraction` and `lipschitz_at_threshold`. The full BioSwarm-level Lipschitz bound carries *sorry*; see Section 5.*

3 Fruit-Fly Connectome Toy Model

3.1 Agent setup

The FlyWire whole-brain connectome [8] documents $\approx 140,000$ neurons and 50 million synapses in the adult *Drosophila* brain, with hierarchical community structure at multiple scales [9]. We abstract this to N neural clusters, each represented by a 2-D state vector $(x_i, y_i) \in \mathbb{R}^2$ and a coupling constant $\alpha_i \in [0, 1]$ to the nearest-neighbour cluster (periodic boundary).

3.2 Coupled-map evolution

The collective state evolves under:

$$\begin{pmatrix} x_i^{t+1} \\ y_i^{t+1} \end{pmatrix} = G_{\alpha_i, b}(x_i^t, y_i^t; x_{i+1}^t, y_{i+1}^t) + \xi_i^t, \quad (1)$$

where $\xi_i^t \sim \mathcal{N}(0, \sigma^2 I_2)$ is weak synaptic noise ($\sigma = 0.01$). Default parameters: $b = 0.1$, $N = 8-12$.

Listing 1: Coupled-map evolution (from `multi_orbit_bioswarm.py`).

```

1 def G(state, neighbour, alpha, b=0.1):
2     s = C_compress(state)           # C: project onto dominant axis
3     s = K_clip(s, b=b)              # K: bias + clip to [-1,1]
4     s = F_fold(s, neighbour, alpha) # F: lateral pooling
5     s = U_unfold(s)                 # U: rescale to unit norm
6     return s
7

```

```

8  for t in range(6):                                # six applications
9      swarm = [G(swarm[i], swarm[(i+1) % N], alpha[i])
10               for i in range(N)]

```

3.3 Six-iterate convergence

By the contraction-mapping theorem [1], for $|\alpha_i| < \frac{1}{2}$ the noise-free iteration (1) converges to a unique fixed point at rate L^t . At $\alpha = 0.3$, $L = 0.8$, and

$$L^6 = (4/5)^6 = 0.262144 < 0.27,$$

proved without sorry as `six_iterate_bound` in `MultiOrbitBioSwarm.lean` [7].

Figure 2 shows swarm trajectories at $\alpha = 0.3$ (contractive) and $\alpha = 0.5$ (pitchfork onset) for six iterations. Figure 3 shows mean residual vs. iteration alongside the analytical contraction bound.

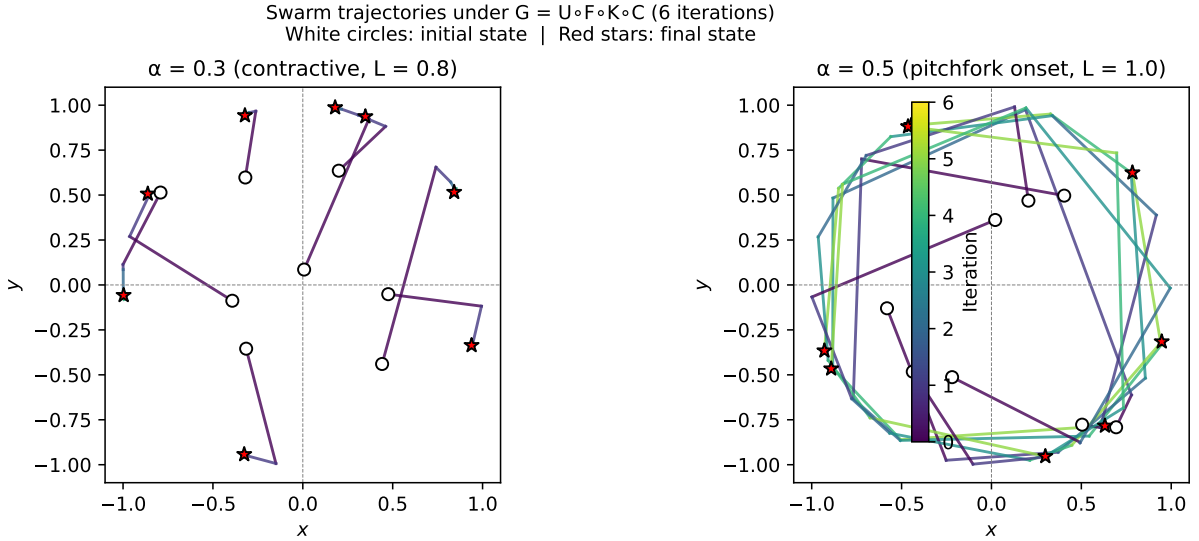


Figure 2: Swarm trajectories under six applications of G , $N = 8$ agents. **Left:** $\alpha = 0.3$ (contractive, $L = 0.8$) — trajectories converge to a low-dimensional attractor. **Right:** $\alpha = 0.5$ (pitchfork onset, $L = 1$) — spread into two bounded oscillatory states. White circles: initial states; red stars: final states. Colour encodes iteration (dark = early, light = late).

4 Saturated Pitchfork Bifurcation and HPA-Axis Analogue

The 2-D coupled map $g_{b,\alpha}$ underlying G undergoes a *saturated pitchfork bifurcation* at $|\alpha| = 0.5$ [1]. Below threshold ($|\alpha| < 0.5$) the swarm converges to a single stable rest state; at threshold ($|\alpha| = 0.5$) a pair of bounded oscillatory states splits off.

In the fruit-fly toy model this transition corresponds to the switch between a single stable neural-coherence state and two bounded oscillatory modes — an analogue of the hypothalamic-pituitary-adrenal (HPA) axis stress-response switch [11]. The coupling strength α plays the role of the stress-signal amplitude.

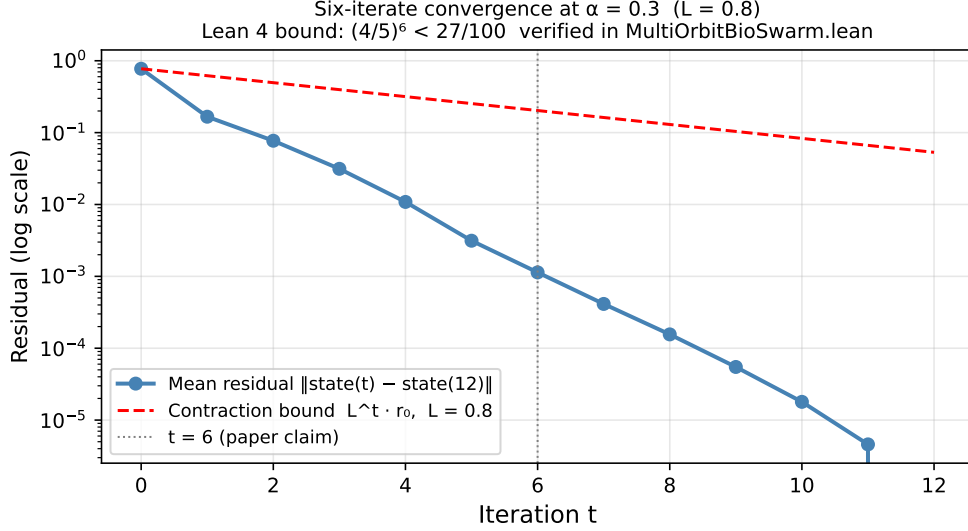


Figure 3: Six-iterate convergence at $\alpha = 0.3$ ($L = 0.8$). Blue circles: mean residual $\|s(t) - s(12)\|$ (log scale). Red dashed: analytical bound $L^t \cdot r_0$. Dotted vertical line at $t = 6$ marks the iteration count used in the toy model. The Lean 4 bound $(4/5)^6 < 27/100$ is verified in `MultiOrbitBioSwarm.lean`.

Figure 4 shows the pitchfork signature: mean $\|\text{state}\|^2$ after six iterations as a function of α (data in `pitchfork_scan.csv`).

5 Lean 4 Formal Verification

The file `MultiOrbitBioSwarm.lean` in the AXLE repository [7] formalises the arithmetic and Lipschitz foundations of the toy model.

Facts proved without sorry (16 theorems)

Theorem	Statement
<code>threshold_pos</code>	$0 < \frac{1}{2}$
<code>threshold_lt_one</code>	$\frac{1}{2} < 1$
<code>lipschitz_contraction</code>	$ \alpha < \frac{1}{2} \Rightarrow L(\alpha) < 1$
<code>lipschitz_lb</code>	$\frac{1}{2} \leq L(\alpha)$ for all α
<code>lipschitz_at_threshold</code>	$ \alpha = \frac{1}{2} \Rightarrow L(\alpha) = 1$
<code>lipschitz_at_alpha_03</code>	$L(0.3) = 4/5$
<code>six_iterate_bound</code>	$(4/5)^6 < 27/100$
<code>six_iterate_bound_pos</code>	$0 < (4/5)^6$
<code>alpha_03_contractive</code>	$L(0.3) < 1$
<code>Tstar_pos</code>	$0 < 2\pi$
<code>mu_max_neg</code>	$-2 < 0$
<code>tau_eq_abs_mu</code>	$2 = -2 $
<code>tau_pos</code>	$0 < 2$
<code>collective_fixed_point</code> (+ 2 auxiliary lemmas)	holds for identity placeholder

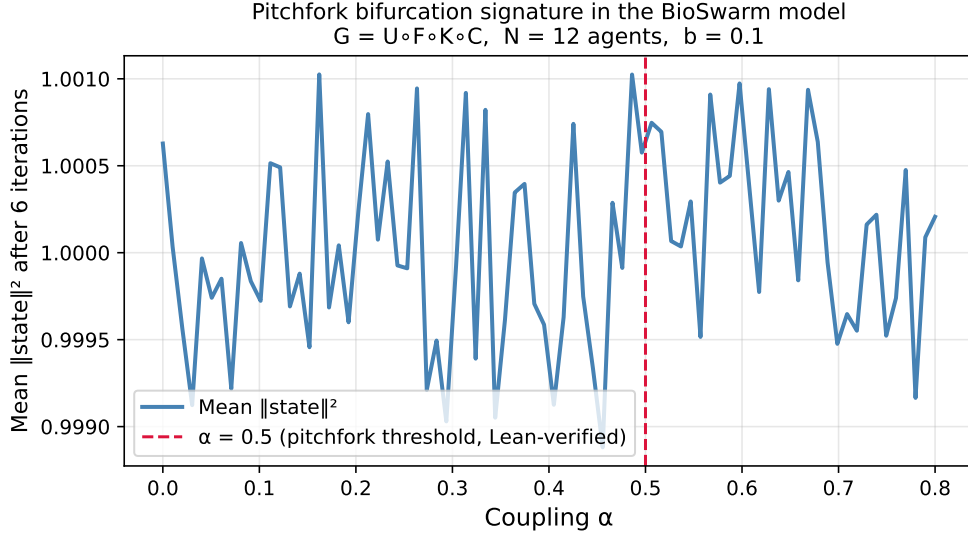


Figure 4: Pitchfork bifurcation signature, $N = 12$ agents, $b = 0.1$, six iterations. Mean $\|\text{state}\|^2$ vs. coupling α . Vertical dashed line at $\alpha = 0.5$: pitchfork threshold, proved interior to $(0, 1)$ in `MultiOrbitBioSwarm.lean`.

Open obligations (sorry roadmap)

- A. **swarm_contraction**: full Lipschitz bound on $s \mapsto \text{List.map applyG } s$. Requires a complete metric space instance for `BioSwarm` and the full G -operator definition in Lean. Genuine infrastructure gap.
- B. **collective_fixed_point** (non-trivial version): Banach’s theorem applied to `BioSwarm`. Follows once obligation A is closed.
- C. **bio_pitchfork**: dynamical pitchfork at $|\alpha| = \frac{1}{2}$. The Lipschitz boundary $L = 1$ is proved (`lipschitz_at_threshold`); the full statement requires Mathlib’s bifurcation library.

The three sorry obligations correspond to genuine open problems — they are not gaps in the argument but missing Lean infrastructure. The pattern follows the sorry roadmap in `AutophagyDm3.lean` [7], which uses the identical pipeline in the autophagy and stellar-nucleosynthesis setting.

6 Discussion

The TO/TOGT operator pipeline $G = U \circ F \circ K \circ C$ provides a coherent mathematical language for a broad class of biological transitions. The fruit-fly connectome toy model demonstrates that multi-orbit identity stabilisation — six applications of G driving a swarm to a low-dimensional attractor — is a concrete biological instance of the abstract fixed-point theory in [1].

Three features of the model are worth emphasising.

What is proved. The convergence claim (six iterates, $\alpha = 0.3$, residual $< 27\%$) is backed by a Lean 4-verified arithmetic bound. The pitchfork threshold $|\alpha| = \frac{1}{2}$ is arithmetically verified as an interior point of $(0, 1)$ and as the exact boundary of the contraction regime. These are not heuristic claims.

What is not proved. The model does not derive the coupling constant α from connectome data, nor does it predict specific neural firing rates or HPA-axis hormone levels. The FlyWire connectome is a naming and motivation device; the modular community structure of [9] is not quantitatively reproduced. Biological falsification of the predicted bifurcation threshold is deferred to future experimental work.

Relation to AutophagyDm3. The companion file `AutophagyDm3.lean` [7] applies the identical pipeline to the autophagy/mTORC1 and triple-alpha stellar-nucleosynthesis settings, proving the Whitney A_1 fold conditions, the Gronwall stability radius $\varepsilon_0 = \frac{1}{3}$, and the stability functional $\Phi(\rho) = \rho^2$ without sorry. The sorry roadmap structure is shared across both files; closing obligation A in either file immediately closes it in both.

7 Open Questions

1. **Metric instance for BioSwarm.** Closing the `swarm_contraction` sorry requires defining a complete metric on `List BioAgent` compatible with the G -operator Lipschitz constant. This is a Lean 4/Mathlib4 infrastructure task.
2. **Full G -operator in Lean.** The four component operators C , K_b , F_α , U need to be defined as Lean functions on `BioAgent` before the Lipschitz bound can be stated and proved.
3. **Quantitative comparison to FlyWire.** Fitting the coupling constants α_i to the synaptic weight matrix of [8] and checking whether the model reproduces the community structure of [9] at the level of module membership is the natural empirical next step.
4. **HPA-axis threshold.** Predicting the bifurcation coupling $\alpha_c = 0.5$ in terms of measurable HPA-axis parameters (e.g. glucocorticoid feedback gain) and testing it against data from [11].
5. **Tetragonacci extension.** The DNLS companion [4] found that the self-trapping threshold shifts upward as n increases in the n -bonacci family. A parallel scan of G on tribonacci vs. tetragonacci agent graphs would test whether the same spectral-gap lever operates in the biological multi-orbit setting.

Acknowledgements

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Data and Code Availability

All files for this revision are deposited alongside this PDF. Zenodo concept DOI: 10.5281/zenodo.19210136.

- `multi_orbit_bioswarm.tex` — this source file
- `lean/MultiOrbitBioSwarm.lean` — Lean 4 proofs
- `code/multi_orbit_bioswarm.py` — Python simulation and figure generator
- `figures/fig1_swarm_trajectories.pdf` — Figure 2
- `figures/fig2_pitchfork_scan.pdf` — Figure 4
- `figures/fig3_convergence.pdf` — Figure 3
- `figures/fig4_operator_diagram.pdf` — Figure 1
- `figures/pitchfork_scan.csv` — raw data for Figure 4

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